



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHRI ASHTALAKSHMI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

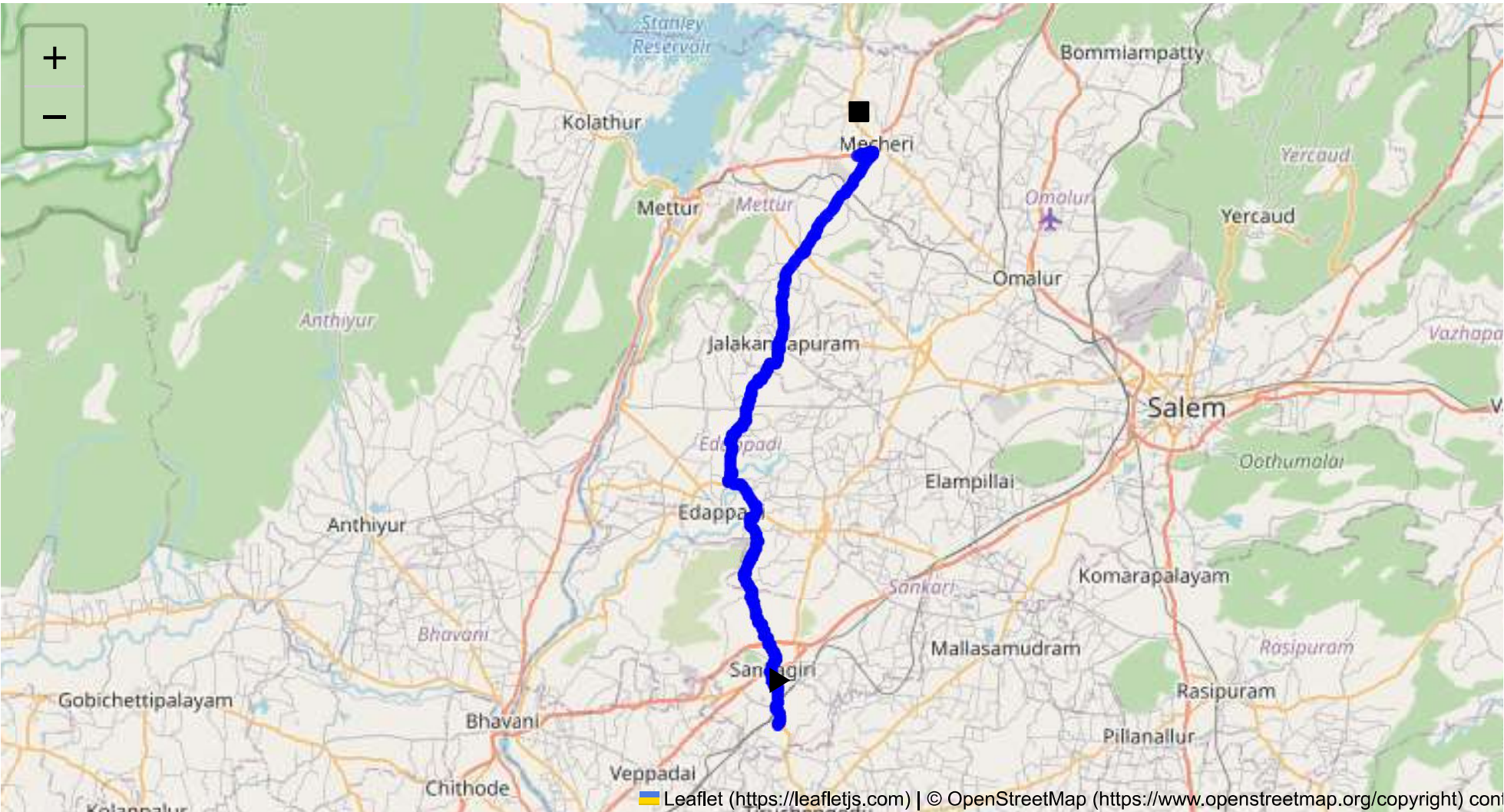
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 52.16 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.82624, 77.9289)



Welcome to the Journey Risk Management Study

Overview of the Route Map

The route from CVQF+23W, Sangagiri, Tamil Nadu 637301 to RWGH+FCF, M. Kalipatti, Tamil Nadu 636453 predominantly involves traveling along regional roads, which can include stretches of National Highway 44 (NH 44). The journey spans approximately 52.16 kilometers, typically requiring about 1.38 hours for heavy vehicles due to speed limits, road conditions, and traffic.

Typical Weather Conditions and Potential Weather-Related Hazards

- **Climate:** The region experiences a tropical climate with hot summers, mild winters, and monsoon rains.
- **Hazards:** The monsoon season (June to September) can bring heavy rainfall leading to water-logging, reduced visibility, and slippery road conditions. Summers can cause high temperatures that may affect vehicle performance.

Analysis of Traffic Patterns

- **Peak Hours:** Significant traffic congestion during morning (7:30 AM - 9:30 AM) and evening (5:30 PM - 7:30 PM) peak hours, particularly near urban areas and major intersections.
- **Congestion-Prone Areas:** Near towns and junctions, especially in Sangagiri and as you approach M.Kalipatti.

Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally variable, with NH 44 being well-maintained but regional roads possibly having potholes and uneven surfaces. Caution is advised on lesser-used roads.
- **Infrastructure:** Adequate signage and lighting on major highways but may be lacking on rural roads.

Suggestions for Alternative Routes for Emergencies

- Detour options via NH 544 if NH 44 is inaccessible, although this may extend travel time. Other regional roads can also be used depending on the specific emergency situation.

Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Transporting hazardous materials requires proper permits and adherence to regulations concerning the types of materials, designated routes, and safety measures.
- **Timing Restrictions:** Some areas may impose time-based restrictions for hazardous material carriers to minimize interaction with peak traffic.

Overview of Historical Incidents

- Recorded incidents involving heavy vehicles are relatively low, with occasional accidents typically due to poor road conditions or driver error.
- Hazardous materials have specific protocols, reducing the frequency of major incidents.

Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Pass through agricultural lands and small settlements; care needs to be taken to minimize air and noise pollution.

- **Wildlife:** Possible crossings in rural stretches, especially near forested areas or water bodies.

Analysis of Communication Coverage

- **Coverage:** Generally good near urban centers and along NH 44. Coverage may be inconsistent in rural stretches and less-populated regions.
- **Dead Zones:** Potentially in isolated areas and forested parts, although these are usually brief.

Estimated Emergency Response Times

- **Urban Areas:** Faster response times (within 20-30 minutes) due to proximity to emergency services.
- **Rural Areas:** Slower response times could range from 45 minutes to over an hour, requiring robust emergency planning.

Overall Summary of Risk Assessment

The route is manageable with moderate risks mainly from traffic conditions, weather, and road quality factors. During the monsoon season, extra caution is needed due to potential flooding and poor visibility. Proper planning, awareness of congestion-prone and low coverage areas, and adherence to regulations can mitigate the majority of risks, ensuring a smoother and safer transit. Emergency response planning is critical, especially for stretches through rural areas. Regular updates on route conditions and alternative pathways are advised for drivers handling hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45000, 77.87213	30 KM/Hr
7	Turn	High	11.45028, 77.87202	15 KM/Hr
8	Turn	Medium	11.45309, 77.87325	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.47451, 77.86914	15 KM/Hr
11	Turn	High	11.47920, 77.86989	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.48070, 77.87166	30 KM/Hr
13	Turn	Medium	11.48155, 77.87174	30 KM/Hr
14	Turn	Medium	11.48277, 77.87250	30 KM/Hr
15	Turn	Medium	11.49496, 77.86660	30 KM/Hr
16	Turn	Medium	11.52571, 77.85522	30 KM/Hr
17	Turn	Medium	11.53598, 77.85142	30 KM/Hr
18	Turn	Medium	11.53857, 77.85087	30 KM/Hr
19	Turn	Medium	11.55027, 77.85506	30 KM/Hr
20	Turn	Medium	11.55460, 77.85624	30 KM/Hr
21	Turn	High	11.57893, 77.85430	15 KM/Hr
22	Turn	Medium	11.57903, 77.85441	30 KM/Hr
23	Turn	Medium	11.60960, 77.84159	30 KM/Hr
24	Turn	Medium	11.61795, 77.84020	30 KM/Hr
25	Turn	Medium	11.62066, 77.84105	30 KM/Hr
26	Turn	Medium	11.64055, 77.84600	30 KM/Hr
27	Turn	Medium	11.64173, 77.84878	30 KM/Hr
28	Turn	Medium	11.64522, 77.85113	30 KM/Hr
29	Turn	Medium	11.66451, 77.85595	30 KM/Hr
30	Turn	Medium	11.67322, 77.86045	30 KM/Hr
31	Turn	Medium	11.67368, 77.86223	30 KM/Hr
32	Turn	Medium	11.68196, 77.86690	30 KM/Hr
33	Turn	High	11.68488, 77.86889	15 KM/Hr
34	Turn	Medium	11.68463, 77.87262	30 KM/Hr
35	Turn	Medium	11.68486, 77.87288	30 KM/Hr
36	Turn	Medium	11.69289, 77.87444	30 KM/Hr
37	Blind Spot	Blind Spot	11.69469, 77.87589	10 KM/Hr
38	Turn	High	11.69491, 77.87546	15 KM/Hr
39	Turn	High	11.69501, 77.87546	15 KM/Hr
40	Turn	High	11.69551, 77.87442	15 KM/Hr
41	Turn	Medium	11.69908, 77.87472	30 KM/Hr
42	Turn	Medium	11.69934, 77.87500	30 KM/Hr
43	Turn	Medium	11.76019, 77.89047	30 KM/Hr
44	Turn	Medium	11.75964, 77.89200	30 KM/Hr
45	Blind Spot	Blind Spot	11.75947, 77.89214	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Blind Spot	Blind Spot	11.82791, 77.94044	10 KM/Hr
47	Turn	Medium	11.82810, 77.94014	30 KM/Hr
48	Turn	High	11.82769, 77.93262	15 KM/Hr
49	Turn	Medium	11.82653, 77.93175	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
9	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
10	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
11	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
12	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
13	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47451, 77.86914



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47920, 77.86989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48070, 77.87166



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48155, 77.87174



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48277, 77.87250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49496, 77.86660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53598, 77.85142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53857, 77.85087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55027, 77.85506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57893, 77.85430



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60960, 77.84159



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61795, 77.84020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62066, 77.84105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64055, 77.84600



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64173, 77.84878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64522, 77.85113



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66451, 77.85595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67322, 77.86045



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67368, 77.86223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68488, 77.86889



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68486, 77.87288



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69289, 77.87444



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69469, 77.87589



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69491, 77.87546



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69501, 77.87546



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69551, 77.87442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69908, 77.87472



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69934, 77.87500



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76019, 77.89047



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75964, 77.89200



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.75947, 77.89214



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.82791, 77.94044



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.82810, 77.94014



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.82769, 77.93262



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.82653, 77.93175